easily. The same goes for time, and our long, plodding second. In the end, the reason that Planck's constant is so small is that we are so big and heavy and slow.

Physicists who are interested in the microscopic world are likely to use units that are more tailored to the phenomena that they study. If we used atomic length scales, time scales, and mass scales, then Planck's constant would not be such an unwieldy number; it would be much closer to 1. In fact, units for which Planck's constant equals 1 are a natural choice for quantum mechanics, and it's a common practice to use them. However, in this book, we will usually retain $\hbar$ in our equations.

4.7 EXPECTATION VALUES

Let's take a short break to discuss an important aspect of statistics, namely the idea of an average value or mean value. We mentioned this idea briefly in the previous lecture, but now it's time to take a closer look.

In quantum mechanics, average values are called expectation values. (In some ways, this is a poor choice of words; I'll tell you why later.) Suppose we have a probability function for the outcome of an experiment that measures an observable **L**. The outcome must be one of **L**'s eigenvalues, λ_i, and the probability function is $P(\lambda_i)$. In statistics, that average (or *mean*) value is denoted by a bar over the quantity being measured. The average of the observable **L** would be $\bar{\mathbf{L}}$. In quantum mechanics, the standard notation is different, having grown out of Paul Dirac's clever bra-ket notation. We represent the average value of **L** with the notation <L>. We'll soon see why the bra-ket notation is so natural, but first let's discuss the meaning of the term *average*.

From a mathematical point of view, an average is defined by the equation

$$\langle \mathbf{L} \rangle = \sum_i \lambda_i P(\lambda_i). \qquad (4.11)$$